Magnetic Resonance Materials in Physics, Biology and Medicine (2024) 37:845–857
https://doi.org/10.1007/s10334-024-01178-3

RESEARCH ARTICLE

s²MRI-ADNet: an interpretable deep learning framework integrating Euclidean-graph representations of Alzheimer's disease solely from structural MRI

Zhiwei Song[1] · Honglun Li[2] · Yiyu Zhang[1] · Chuanzhen Zhu[1] · Minbo Jiang[1] · Limei Song[3] · Yi Wang[1,6] · Minhui Ouyang[4,5] · Fang Hu[6] · Qiang Zheng[1]

Received: 6 March 2024 / Revised: 19 May 2024 / Accepted: 4 June 2024 / Published online: 13 June 2024
© The Author(s), under exclusive licence to European Society for Magnetic Resonance in Medicine and Biology (ESMRMB) 2024

Abstract

Objective To establish a multi-dimensional representation solely on structural MRI (sMRI) for early diagnosis of AD.

Methods A total of 3377 participants' sMRI from four independent databases were retrospectively identified to construct an interpretable deep learning model that integrated multi-dimensional representations of AD solely on sMRI (called s²MRI-ADNet) by a dual-channel learning strategy of gray matter volume (GMV) from Euclidean space and the regional radiomics similarity network (R2SN) from graph space. Specifically, the GMV feature map learning channel (called GMV-Channel) was to take into consideration spatial information of both long-range spatial relations and detailed localization information, while the node feature and connectivity strength learning channel (called NFCS-Channel) was to characterize the graph-structured R2SN network by a separable learning strategy.

Results The s²MRI-ADNet achieved a superior classification accuracy of 92.1% and 91.4% under intra-database and inter-database cross-validation. The GMV-Channel and NFCS-Channel captured complementary group-discriminative brain regions, revealing a complementary interpretation of the multi-dimensional representation of brain structure in Euclidean and graph spaces respectively. Besides, the generalizable and reproducible interpretation of the multi-dimensional representation in capturing complementary group-discriminative brain regions revealed a significant correlation between the four independent databases ($p < 0.05$). Significant associations ($p < 0.05$) between attention scores and brain abnormality, between classification scores and clinical measure of cognitive ability, CSF biomarker, metabolism, and genetic risk score also provided solid neurobiological interpretation.

Conclusion The s²MRI-ADNet solely on sMRI could leverage the complementary multi-dimensional representations of AD in Euclidean and graph spaces, and achieved superior performance in the early diagnosis of AD, facilitating its potential in both clinical translation and popularization.

Keywords Alzheimer's disease · Structural MRI · Multi-dimensional representations · Deep learning · Interpretability

Zhiwei Song and Honglun Li are the equally contributing first authors.

✉ Qiang Zheng
zhengqiang@ytu.edu.cn

[1] School of Computer and Control Engineering, Yantai University, Yantai 264005, China

[2] Department of Radiology, Yantai Yuhuangding Hospital Affiliated with Qingdao University Medical College, Yantai 264099, China

[3] School of Medical Imaging, Weifang Medical University, Weifang 261000, China

[4] Department of Radiology, Children's Hospital of Philadelphia, Philadelphia, PA 19104, USA

[5] Department of Radiology, Perelman School of Medicine, University of Pennsylvania, Philadelphia, PA 19104, USA

[6] Key Laboratory of Medical Imaging and Artificial Intelligence of Hunan Province, Xiangnan University, Chenzhou 423000, Hunan, China

Introduction

Structural magnetic resonance imaging (sMRI) particularly for T1-weighted image (T1WI) is the most clinically accessible brain imaging technology and has been widely studied in the early diagnosis of Alzheimer's disease (AD) [1]. With the development of the advanced MRI sequences, such as functional MRI (fMRI) [2], diffusion tensor imaging (DTI) [3], and arterial spin labeling (ASL) [4] etc., a large number of studies on multi-modality or on single-modality but with advanced MRI sequence are carried out for early diagnosis of AD [5]. The investigation of the advanced sequences in AD is beneficial because multi-modality MRIs are able to uncover brain structure and function from different perspectives [6]. However, the advanced MRI sequences are not always clinically available in a considerable number of hospitals, which hinders its translation and popularization. Therefore, a well-performing model for early diagnosis of AD established solely on sMRI is of great importance, but needs to probe and integrate a multi-dimensional representation of AD using advanced deep learning techniques.

The convolutional neural network (CNN) conducted on gray matter volume (GMV) feature map of specific brain region such as hippocampus [7] or the whole brain [8] is a typical strategy established on T1WI for early diagnosis of AD because GMV reduction is a prominent feature that related to cognitive impairment. However, the CNN struggles with aggregating long-range spatial dependencies in images, and the GMV, being a feature map in Euclidean space, overlooks features with non-Euclidean structures in the brain network.

Regarding the limitations of CNN, Transformer has emerged as a powerful approach to characterize the global contextual information by employing a multi-head self-attention mechanism [9], and the CNN-Transformer is expected to learn more representative features of AD by taking into consideration spatial information of both long-range spatial relations and detailed localization information in images [10].

Regarding the non-Euclidean structure observed in brain networks derived from conventional T1WI, there has been growing interest in morphological brain network analysis. This approach calculates the covariance between brain regions using morphometric features, such as regional cortical thickness and voxel intensity [11]. Particularly, the regional radiomics similarity network (R2SN) which is conducted on radiomics features [12] has been demonstrated a powerful approach in AD analysis with robustness and biological basis, such as revealing distinct subtypes and abnormality patterns in mild cognitive impairment (MCI) [13]. In addition, the advanced graph neural network (GNN) is a powerful tool for characterizing unstructured and non-Euclidean data, and thus has been widely used to represent and analyze the brain fMRI [14] and DTI [15] networks. However, there is still a literature gap regarding leveraging the brain connectivity structure and radiomics properties of the morphological brain network R2SN by GNN.

The aim of the study was to establish a multi-dimensional representation solely on sMRI for early diagnosis of AD, and thus an interpretable deep learning model (called s^2MRI-ADNet) was proposed by a dual-channel learning of gray matter volume (GMV) from Euclidean space and the regional radiomics similarity network (R2SN) from graph space (Fig. 1). The complementary, generalizable, reproducible, and the neuroscientific interpretation of the multi-dimensional representation of s^2MRI-ADNet were also validated on 3377 subjects from four publically available databases.

Materials and methods

Participants

In our study, we selected 3377 subjects from the Alzheimer's Disease Neuroimaging Initiative database (ADNI) (http://adni.loni.usc.edu), Australian Imaging Biomarkers and Lifestyle (AIBL) (http://aibl.csiro.au), the European DTI Study on Dementia database (EDSD) (http://neugrid4you.eu) and the Open Access Series of Imaging Studies (OASIS) (http://oasis-brains.org) (Table 1 and supplementary material S01). Of note, the ADNI database was used as the discovery cohort because of its abundant clinical measures, including cognitive ability (MMSE, ADAS11, ADAS13, ADASQ4, CDRSB, FAQ, Ravlt-immediate (AVLT-1) and Ravlt-learning (AVLT-2)), CSF biomarker (CSF Aβ, CSF Tau, and CSF P-Tau), metabolism (FDG), and genetic risk score (PHS).

Data preprocessing and R2SN construction

For each individual's T1WI scan, the GMV map of the whole-brain was segmented out via the CAT12 toolkit (http://dbm.neuro.uni-jena.de/cat/) with the standardized procedures. Further, all sMRI data were bias-corrected, segmented into gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF) and registered to Montreal Neurological Institute (MNI) space using a sequential linear (affine) transformation. The gray matter image ($181\times217\times181$) was resliced into $91\times109\times91$ with 2 mm^3 isotropic voxel before feeding into the deep learning model, aiming to alleviate the computational complexity and memory limitation.

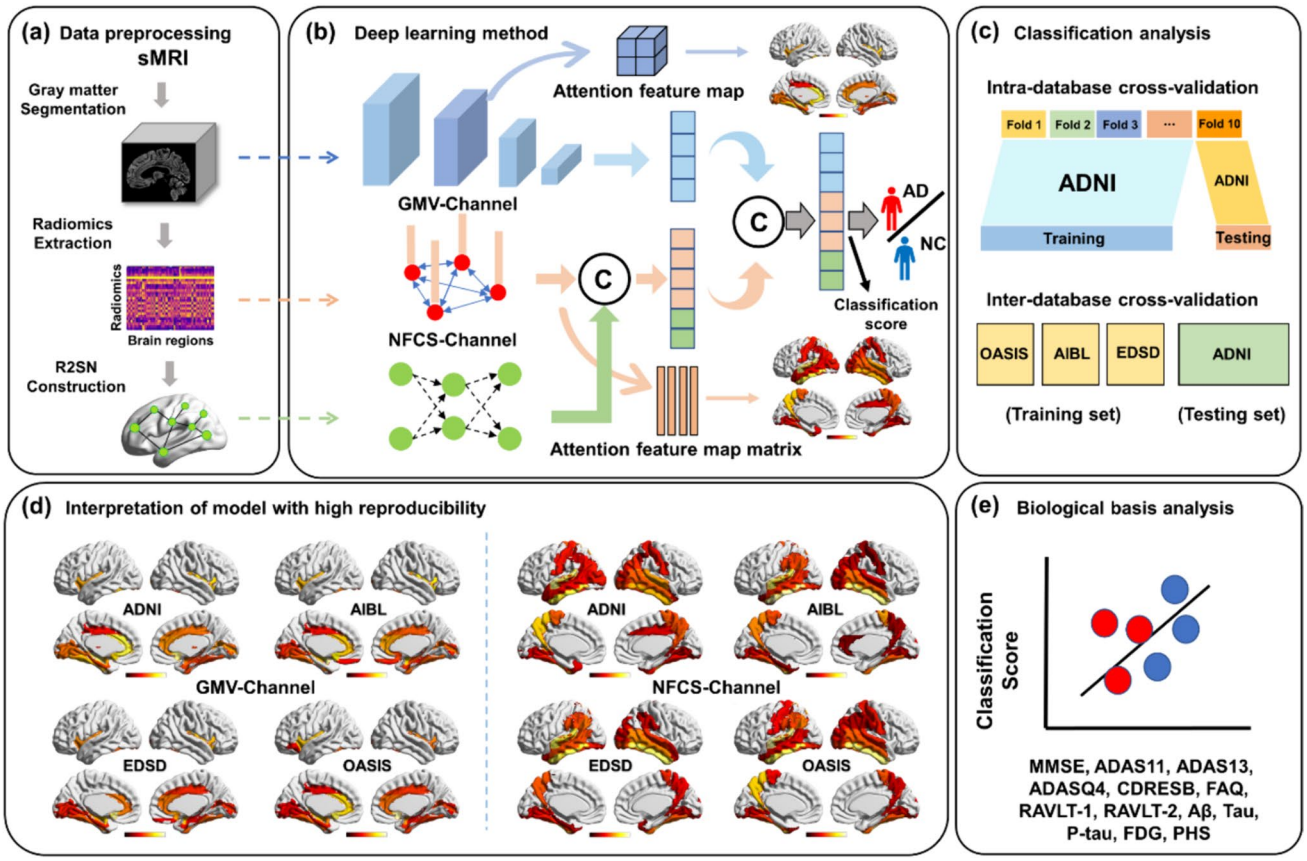


Fig. 1 Schematic illustration of the method pipeline. **a** Pipeline of the data preprocessing. **b** The proposed s^{2}MRI-ADNet multi-dimensional representations of spatial information and brain connectivity. **c** The intra-database cross-validation and inter-database cross-validation strategies. **d** Visualization of important brain regions captured by the dual-channel of the proposed s^{2}MRI-ADNet. **e** Correlation analysis of classification scores and clinical measures. *SCN* structure connectivity network

Table 1 Demographic information about the subjects involved in four independent databases of ADNI, AIBL, EDSD, and OASIS.

Cohort	Group	Age (years)	Sex (M/F)	MMSE
ADNI ($N=1649$)	NC (603)	73.46 ± 6.16	277/326	29.08 ± 1.10
	MCI (764)	72.98 ± 7.68	447/317	27.56 ± 1.81
	AD (282)	74.91 ± 7.69	151/131	23.18 ± 2.13
	p value	< 0.001	< 0.001	< 0.001
AIBL ($N=557$)	NC (478)	73.10 ± 6.10	203/275	28.70 ± 1.24
	AD (79)	74.16 ± 7.84	33/46	20.41 ± 5.49
	p value	0.172	0.908	< 0.001
EDSD ($N=388$)	NC (230)	68.76 ± 6.14	108/112	28.58 ± 2.97
	AD (158)	75.54 ± 8.10	66/92	20.89 ± 5.12
	p value	< 0.001	0.072	< 0.001
OASIS ($N=783$)	NC (594)	67.07 ± 8.70	240/354	29.06 ± 1.54
	AD (189)	75.04 ± 7.70	95/94	24.47 ± 4.14
	p value	< 0.001	< 0.05	< 0.001

ADNI the Alzheimer's disease neuroimaging initiative database, *AIBL* Australian imaging biomarkers and lifestyle, *EDSD* the European DTI study on dementia database (EDSD), *OASIS* the open access series of imaging studies

In this study, the R2SN was established on the calculated GMV feature map rather than the raw intensity-based T1WI images because GMV is a significant biomarker associated with AD [16] (Fig. 1a). First, a series of well-defined radiomics features ($N=25$), including intensity and texture features, were calculated for each automated anatomical labeling-derived region of interest (AAL-derived ROI) ($N=90$) (https://github.com/YongLiulab). Then, each radiomics feature was normalized across all ROIs capitalizing upon a common min–max method. At last, the R2SN with size of 90×90 for each individual was established by the Pearson correlation coefficient (PCC) between the radiomics features of each pair of ROIs [12].

The proposed s^{2}MRI-ADNet

The proposed s^{2}MRI-ADNet was summarized in Figs. 1b, 2a, including a GMV-Channel characterizing the GMV feature map by a CNN-Transformer network to incorporate both long- and short-ranged contextual information of the whole brain, and a NFCS-Channel characterizing

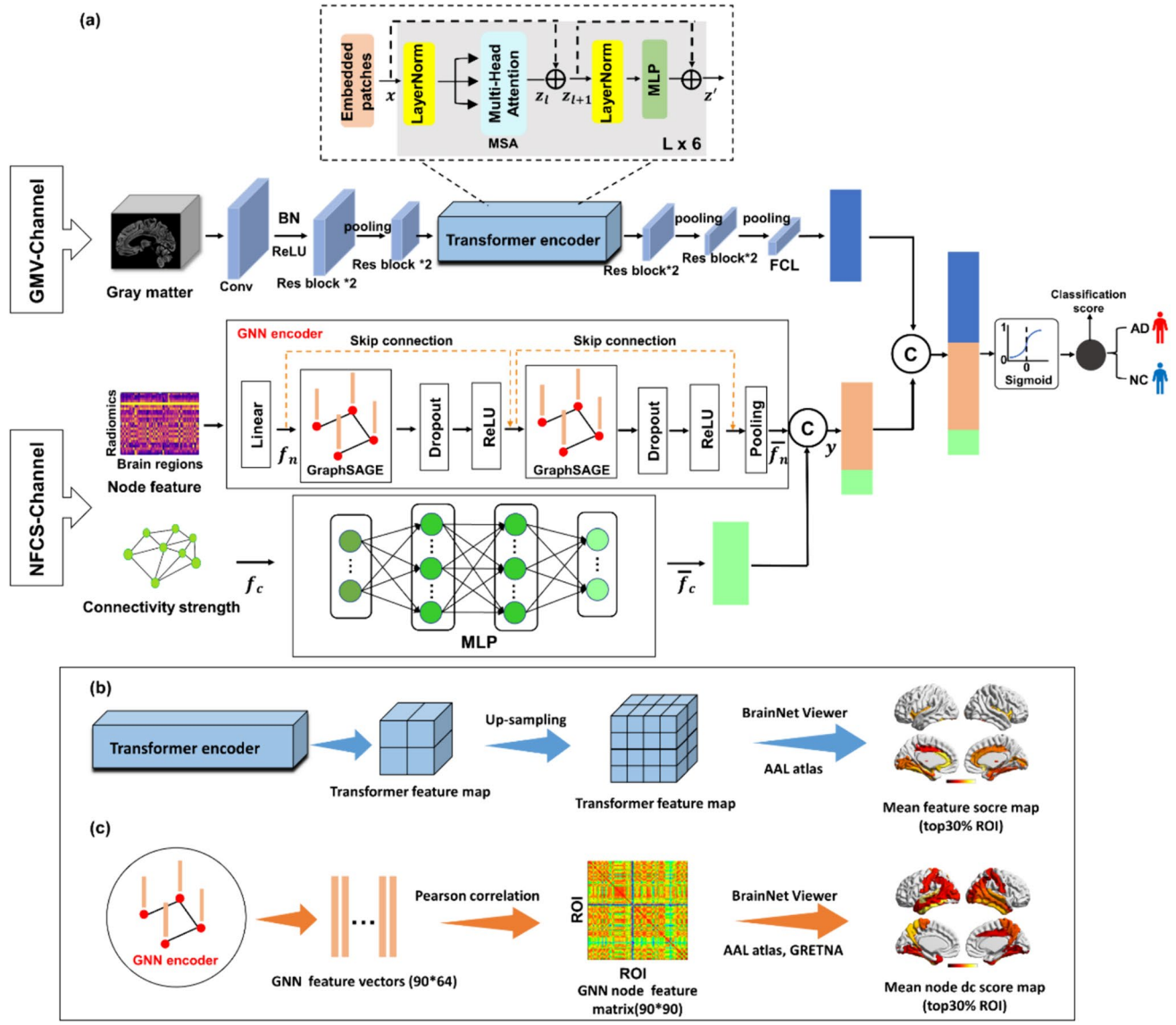


Fig. 2 Architecture and interpretation of the proposed s^{2}MRI-ADNet. **a** The detailed architecture of the dual-channel deep learning model s^{2}MRI-ADNet including GMV-Channel and NFCS-Channel. **b** The interpretable feature map generated by the Transformer encoder module in the GMV-Channel. **c** The interpretable feature map generated by the GNN encoder module in the NFCS-Channel. *GNN* graph neural network

the non-Euclidean structured R2SN network in terms of node radiomics features by GNN technique and connection strength features by MLP technique separately. Finally, the s^{2}MRI-ADNet computed a feature vector x, which was obtained by concatenating the output features of GMV-Channel and NFCS-Channel. This feature vector x was then transformed into a classification probability value between 0 and 1, also known as a classification score, using a sigmoid function.

The GMV feature map learning channel (GMV-Channel)

The GMV-Channel was established by a CNN-Transformer fusion framework (Fig. 2a), which was comprised of a convolution layer, four residual blocks [17] for feature characterization, six Transformer encoder blocks [18] to aggregate image features in terms of long distances, another four residual blocks to enhance the aggregated feature representation generated from the Transformer encoder block, and a fully connected layer. Specifically, the residual block was conducted by two convolutional layers, a batch normalization

layer, a nonlinear activation (ReLU) layer, and a typical skip connection to alleviate the degradation problem [17]. Between the former and latter residual blocks, the Transformer encoder block was embedded due to its better feature aggregation capability over long distances than CNN networks [19].

Concretely, each Transformer encoder block concatenated two residual connection architectures. The first residual connection architecture, which consisted of a LayerNorm (LN) layer [20] and a multi-head self-attention (MSA) module, calculated the feature map $z_{l+1} = MSA(LN(x)) + x$ with x being the embedded patches (Fig. 2a). The Transformer encoder block adopted the multi-head attention mechanism to calculate the attention coefficient between each patch of the input feature map and captured the global information. The second residual connection architecture, which consisted of a LN layer and a MLP layer for a nonlinear mapping, generated the final feature map z' of each Transformer encoder block by determining $z' = MLP\left(LN\left(z_{l+1}\right)\right) + z_{l+1}$ (Fig. 2a).

The node feature and connectivity strength learning channel (NFCS-Channel)

The NFCS-Channel consisted of a node feature learning network conducted by a GNN encoder module and a connectivity strength learning network conducted by a MLP module (Fig. 2a). The GNN encoder was established by two residual GraphSAGE structures, where each structure concatenated a GraphSAGE block [21], a Dropout layer, a ReLU layer, and a typical skip connection. Unlike the conventional GNN that often relies on a fixed aggregation function (such as mean or sum), GraphSAGE introduces a learnable function that enables the model to adaptively sample and aggregate information from neighboring nodes [21]. The adaptive aggregation mechanism of GraphSAGE is able to effectively generalize representations for nodes in a graph by aggregating information from their local neighborhoods [22]. Besides, a linear layer and a global average pooling layer were added before and after the two residual GraphSAGE structures to avoid over smoothing and to aggregate the node embedding into the graph embedding, respectively. The MLP module consisted of an input layer, two hidden layers and an output layer. Each hidden layer was followed by a nonlinear activation function ReLU and a dropout layer to avoid over fitting.

The GraphSAGE adopted in the GNN encoder module uses node feature information to generate node embeddings on unseen nodes or graphs, and thus does not support weighted graphs. Therefore, the MLP module was adopted as a complementary strategy to learn the connectivity strength information of the graph. Since the values in the PCC matrix of R2SN graph represent the connectivity strengths between the radiomics features of each pair of brain ROIs [12], we used the PCC matrix to generate inputs for MLP module. The separable learning strategy is beneficial by taking into account both node feature f_n and connectivity strength feature f_c. Specifically, the GNN encoder module was fed with the topological structure of R2SN and the radiomics features of each brain region for the node feature learning, generating $\overline{f_n} = AvgPool[\sigma(GraphSAGE(f_n))]$, where σ is the ReLU function. The MLP module was fed with a feature vector where each component was from the upper triangular of the PCC matrix of R2SN, generating $\overline{f_c} = MLP(f_c)$. Of note, both the topological structure and the upper triangular matrix were obtained by applying a threshold 0.8 on the PCC matrix of R2SN (The determination of the threshold was summarized in the Supplementary Table S1).

Implementation of the s²MRI-ADNet

The experiments were implemented on Pytorch 1.7.1. The Binary CrossEntropy (BCE) was employed as loss function, and the hyperparameters were optimized with Adam optimizer with a learning rate equal to 3×10^{-6} and a weight decay rate equal to 1×10^{-6}. The batch size was set to 2 to fit the GPU memory. The model was trained for 100 epochs. All the codes of s²MRI-ADNet have been released via GitHub (https://github.com/FindBrain/s2MRI_ADNet).

The complementary, generalizable, and reproducible interpretation of s²MRI-ADNet based on attention scores

Attention score refers to the weights used in an attention mechanism [23, 24]. Attention mechanisms are employed in feature-based tasks to weight different parts of the input features, allowing the model to focus more on important segments during processing [25]. These attention scores are typically learned by the model and are computed based on the input context and task, guiding the model's attention to specific parts of the input feature [23]. Therefore, attention scores guide the model's focus on specific parts of input features, and determine the importance of the brain regions in the early diagnosis of AD.

The complementary interpretation of the s²MRI-ADNet was carried out in terms of both GMV-Channel and NFCS-Channel in capturing complementary and significant brain regions (Fig. 2). The generalizable and reproducible interpretation of the multi-dimensional representation in capturing complementary discriminative brain regions was performed by correlating the specific attention-score defined below between the four independent databases.

Regarding the GMV-Channel (Fig. 2b), the feature map was generated by the Transformer encoder module. After an up-sampling operation on the feature map, the average score

of each brain region throughout all the testing data, which constructed the GMV-Channel attention feature map (called GMV-Channel-attention-map), was obtained to determine the importance of the brain regions in the early diagnosis of AD. The calculated score of each brain region on the GMV-Channel-attention-map was called as GMV-Channel-attention-score. In this study, the brain regions with top 30% GMV-Channel-attention scores ($90 \times 0.3 = 27$) were visualized through BrainNet Viewer toolbox [26].

Regarding the NFCS-Channel (Fig. 2c), the GNN encoder module generated a 90×64 feature matrix where 90 and 64 were the brain region (or node number) and node feature vector for each brain region. Then, a PCC matrix (90×90) was calculated based on the generated node feature matrix, respectively. The PCC matrix was averaged throughout all the testing data. The degree centrality of the 90 brain regions, which conducted the NFCS-Channel attention feature map matrix (called NFCS-Channel-attention-map), was calculated on the generated averaged PCC matrix by employing the GRETNA toolbox [27] to determine the importance of the brain regions in the early diagnosis of AD. The calculated degree centrality for all brain regions on the NFCS-Channel-attention-map was called as NFCS-Channel-attention-score. Similarly to the GMV-Channel, the brain regions with top 30% NFCS-Channel-attention scores ($90 \times 0.3 = 27$) were visualized through BrainNetViewer toolbox [26].

The neuroscientific interpretation of s²MRI-ADNet based on classification scores

Classification score refers to the probabilities assigned by a model to different classes for a given input sample in a classification task. In classification, the model assigns a score or probability to each possible class, often using a sigmoid function to convert these scores into a probability distribution [28]. Ultimately, the model selects the class with the highest score as the predicted result. Classification score is widely used method to demonstrate the clinical biological basis of deep learning models [13, 29].

To validate the clinical relevance of the s²MRI-ADNet, we investigated the correlation between the classification score generated by the s²MRI-ADNet and number of clinical measures in ADNI database. The clinical measures included cognitive ability (MMSE, ADAS11, ADAS13, ADASQ4, CDRSB, FAQ, Ravlt-immediate and Ravlt-learning), CSF biomarker (CSF Aβ, CSF Tau, and CSF P-Tau), metabolism (FDG), and genetic risk score (PHS) (Fig. 1e). The correlation analysis was performed in the MCI and AD groups in ADNI database after controlling for the effects of age, gender, and clinical group classes. Besides, the neuroscientific interpretation of s²MRI-ADNet was also carried out by correlating the attention scores and brain abnormality (t statistic score obtained by the statistical comparison between NC vs AD $p < 0.05$).

Results

Demographic characteristics

In this study, a total of 3377 subjects were collected from four independent databases, including 1649 subjects from ADNI, 557 subjects from AIBL, 388 subjects from EDSD, 783 subjects from OASIS. In the ADNI cohort, statistically significant difference in MMSE score was observed between NC, MCI, and AD groups ($p < 0.001$, ANOVA test). In the AIBL, EDSD, OASIS cohorts, statistically significant difference in MMSE score was observed between NC and AD groups ($p < 0.001$, t-test). The summarized demographic information was displayed in Table 1.

Comparison of NC vs AD classification performance

To compare the NC vs AD classification performance, we carried out the experiments by solely feeding deep learning models with GMV feature maps, or non-Euclidean structured R2SN network, or the combination of them. When fed with GMV feature maps, four alternative deep learning models of 3DResNet [30], 3D attention network (3DAN) [29], Vision Transformer (VIT) [18], and Residual Transformer (ResTransformer) [31] were employed for comparison. When fed with R2SN network, four alternative GNN-MLP models of graph convolutional network [32] and MLP [33] (GCN-MLP), graph attention network [34] and MLP (GAT-MLP), graph isomorphism network [35] and MLP (GIN-MLP), graph sample and aggregate [21] and MLP (GraphSage-MLP) were employed for comparison.

When fed with the combination of GMV feature map and R2SN network, the proposed s²MRI-ADNet was adopted. Besides, an intra-database cross-validation (Fig. 1c, Table 2) and an inter-database cross-validation (Fig. 1c, Table 3) were carried out separately in this study.

Table 2 summarized experimental results solely within the ADNI database using a tenfold cross-validation (Fig. 1c). Although solely established on conventional T1WI image, the accuracy (ACC), sensitivity (SEN), specificity (SPE), and the area under the receiver operating characteristic (AUC) of the s²MRI-ADNet reached ACC = 92.1%, SEN = 88.0%, SPE = 94.2%, AUC = 0.962. The diagnostic accuracy of s²MRI-ADNet is about 2–6% higher than solely adopting GMV feature map, and about 8–10% higher than solely adopting the non-Euclidean structured R2SN network.

Table 3 summarized experimental results of the inter-database cross-validation among multi-center databases of ADNI, AIBL, EDSD, and OASIS (Fig. 1c). The ACC, SEN,

Table 2 Comparison of the proposed s^{2}MRI-ADNet with other deep learning-based methods in classifying AD and NC in the ADNI cohort under the intra-database cross-validation.

Feature	Model	ACC (%)	SEN (%)	SPE (%)	AUC
R2SN	GCN-MLP	84.3	72.0	89.7	0.892
	GAT-MLP	83.1	65.0	91.5	0.895
	GIN-MLP	82.6	64.2	91.1	0.889
	GraphSAGE-MLP	82.5	63.6	91.3	0.891
Gray matter volume	3DResNet	85.9	73.3	91.8	0.927
	3DAN	89.9	85.4	92.0	0.947
	VIT	86.0	69.2	93.8	0.916
	ResTransformer	90.1	82.2	93.8	0.959
Combination	s^{2}MRI-ADNet	**92.1**	**88.0**	**94.2**	**0.962**

AD Alzheimer's disease, *NC* normal control, *ACC* accuracy, *SPE* specificity, *SEN* sensitivity, *AUC* the area under the receiver operating characteristic. Best in bold

Table 3 Comparison of the proposed s^{2}MRI-ADNet with other models in classifying AD and NC under the inter-database cross-validation between ADNI, AIBL, EDSD, and OASIS (Best in bold).

Feature	Model	ACC (%)	SEN (%)	SPE (%)	AUC
R2SN	GCN-MLP	84.1	66.3	91.3	0.877
	GAT-MLP	83.7	68.6	89.6	0.871
	GIN-MLP	85.2	64.9	91.4	0.883
	GraphSAGE-MLP	84.5	68.2	90.3	0.872
Gray matter volume	3DResNet	89.1	70.8	95.5	0.930
	3DAN	89.3	73.9	94.5	0.932
	VIT	88.9	72.4	95.5	0.928
	ResTransformer	89.8	76.5	94.5	0.936
Combination	s^{2}MRI-ADNet	**91.4**	**77.2**	**96.4**	**0.938**

AD Alzheimer's disease, *NC* normal control, *ADNI* the Alzheimer's disease neuroimaging initiative database, *AIBL* Australian imaging biomarkers and lifestyle, *EDSD* the European DTI study on dementia database (EDSD), *OASIS* the open access series of imaging studies

SPE, and AUC of the s^{2}MRI-ADNet reached ACC = 91.4%, SEN = 77.2%, SPE = 96.4%, and AUC = 0.938, demonstrating the robustness, reproducibility, and generalizability of the proposed s^{2}MRI-ADNet in early diagnosis of AD when establishing solely on the conventional T1WI images.

Interpretation of the s^{2}MRI-ADNet

Important regions captured by the transformer encoder and GNN encoder were complementary, generalizable, and reproducible

The complementary, generalizable, and reproducible interpretation of s^{2}MRI-ADNet were investigated to reveal the underlying fundamentals in facilitating classification performance (Fig. 1d). Of note, the inter-database cross-validation setting (ADNI, AIBL, EDSD, OASIS) indicated that a brain region set would be identified in each round of inter-database cross-validation. Concretely, when the ADNI was assigned as testing set, the identified brain region set was defined as ADNI-set. Likewise, the AIBL-set, EDSD-set, OASIS-set were obtained accordingly.

The complementary interpretation of s^{2}MRI-ADNet: Fig. 3a, b displayed the discriminative brain regions generated by the GMV-Channel-attention-map and the NFCS-Channel-attention-map based on the inter-database study. The brain regions are also listed in Table 4. The involved brain regions above were the intersection of the four sets of ADNI-set, AIBL-set, EDSD-set, and OASIS-set. From the results, GNN encoder captured the brain regions that the Transformer encoder did not concentrate on. It indicated that the GMV-Channel and the NFCS-Channel integrated in the s^{2}MRI-ADNet extracted complementary spatial information in Euclidean space and brain connectivity in graph space, suggesting a sufficient feature extraction solely from sMRI.

The generalizable and reproducible interpretation of s^{2}MRI-ADNet: Fig. 3c demonstrated the generalizable and reproducible interpretation regarding the GMV-Channel-attention-map. The GMV-Channel-attention-score of the group-discriminative brain regions exhibited a significant correlation between ADNI vs AIBL, ADNI vs EDSD, ADNI vs OASIS, AIBL vs EDSD, AIBL vs OASIS, EDSD vs

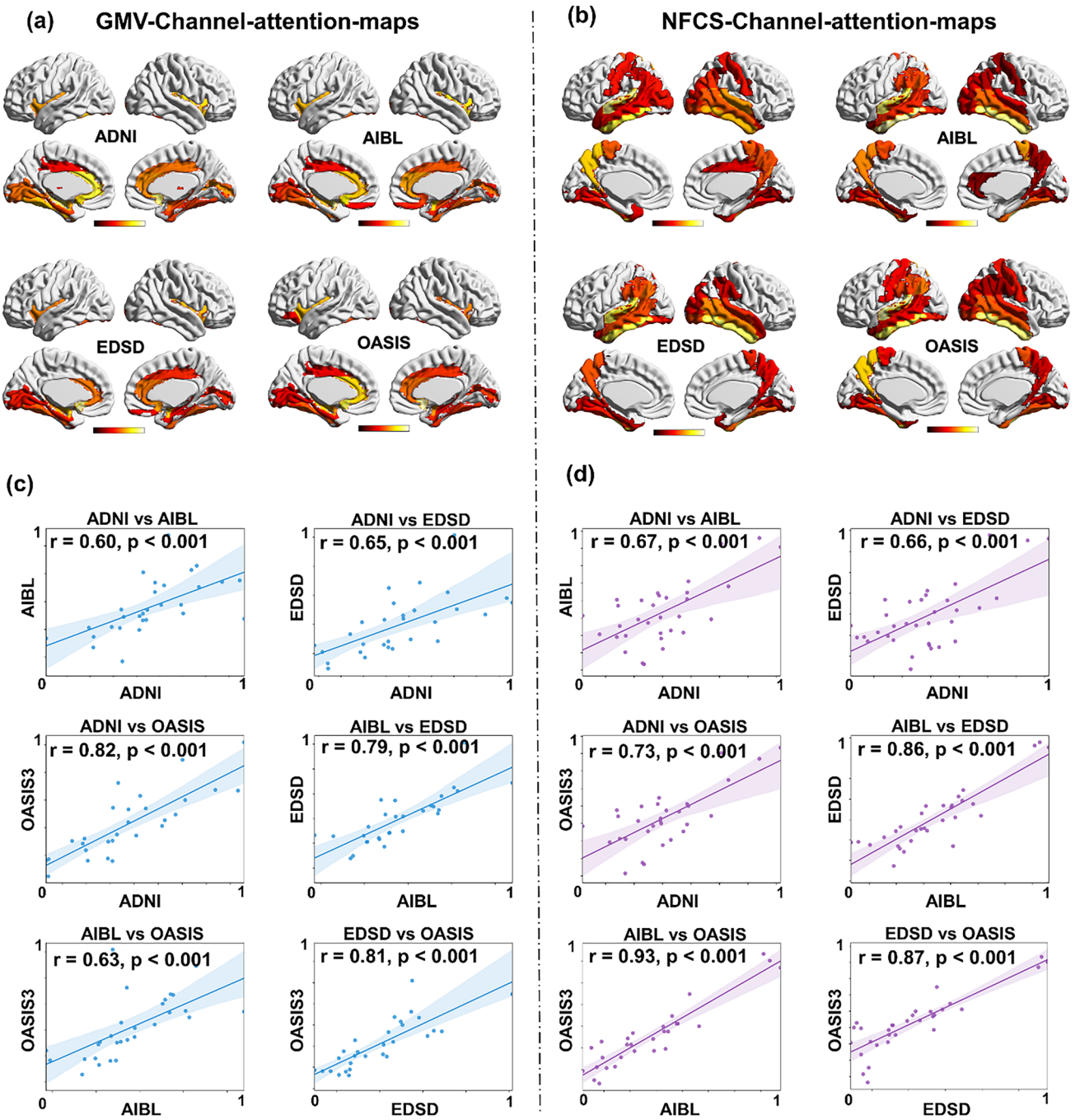


Fig. 3 Important brain regions captured by proposed s^{2}MRI-ADNet and their correlation between four databases of ADNI, AIBL, EDSD, OASIS. **a** GMV-Channel-attention-maps highlighted by the top 30% brain regions on the four databases. **b** NFCS-Channel-attention-maps highlighted by the top 30% brain regions on the four databases. **c** The correlation of the GMV-Channel-attention-maps between the four databases. **d** The correlation of the NFCS-Channel-attention-maps between the four databases. *ADNI* the Alzheimer's disease neuroimaging initiative database, *AIBL* Australian imaging biomarkers and lifestyle, *EDSD* the European DTI study on dementia database (EDSD), *OASIS* the open access series of imaging studies

OASIS, demonstrating the consistency of the GMV-Channel-attention-score under the inter-database setting. In each scatter plot of Fig. 3c, taking (ADNI-set ∪ AIBL-set) for example, the identified brain regions from the union set of ADNI-set and AIBL-set were displayed as blue dots. Likewise, Fig. 3d demonstrated the generalizable and reproducible interpretation regarding the

Table 4 The important brain regions captured by the dual channels of the GMV-Channel and NFCS-Channel in the proposed s^{2}MRI-ADNet

Brain regions (GMV-Channel)	Brain regions (NFCS-Channel)
Olfactory cortex	Lingual gyrus
Insula	Middle occipital gyrus
Cingulate gyrus	Inferior occipital gyrus
Hippocampus	Fusiform gyrus
Parahippocampal gyrus	Angular gyrus
Amygdala	Precuneus
Lingual gyrus	Paracentral lobule
Fusiform gyrus	Superior temporal gyrus
Caudate nucleus	Middle temporal gyrus
Lenticular nucleus	Inferior temporal gyrus
Thalamus	
Transverse temporal gyrus	

NFCS-Channel-attention-map between the four databases of ADNI, AIBL, EDSD, and OASIS.

The representative interpretability of s^{2}MRI-ADNet: We randomly selected 3 individuals from each test database for top 30% brain regions visualization at the individual level using the BrainNet Viewer toolbox [26]. The supplementary material S02 showed that the high-signal regions of GMV-Channel-attention-maps and NFCS-Channel-attention-maps at the individual level were consistent with those at the group level shown in Fig. 3a, b.

The attention maps from the dual-channel of s^{2}MRI-ADNet were significantly associated with brain abnormality

To evaluate whether the GMV-Channel-attention-map derived from the Transformer encoder and NFCS-Channel-attention-map derived from the GNN encoder were associated with brain atrophy in AD analysis, we performed a correlation study between the mean attention scores of all the 90 brain regions and the *t*-values (*t* statistic score) obtained by the statistical comparison between NC vs AD for the 90 ROIs.

Figure 4a displayed the scatter plots between the GMV-Channel-attention-map and the t-values, demonstrating significant correlations in the ADNI database ($r = 0.22$, $p < 0.05$), the AIBL database ($r = 0.27$, $p < 0.05$), the EDSD database ($r = 0.25$, $p < 0.05$) and the OASIS3 ($r = 0.22$, $p < 0.05$). Figure 4b displayed the scatter plots between NFCS-Channel-attention-map and the t-values, demonstrating significant correlations in the ADNI database ($r = 0.28$, $p < 0.01$), the AIBL database ($r = 0.33$, $p < 0.01$), the EDSD database ($r = 0.43$, $p < 0.01$) and the OASIS3 ($r = 0.29$, $p < 0.01$). Both Fig. 4a, b indicated that the proposed s^{2}MRI-ADNet could capture the abnormalities in AD.

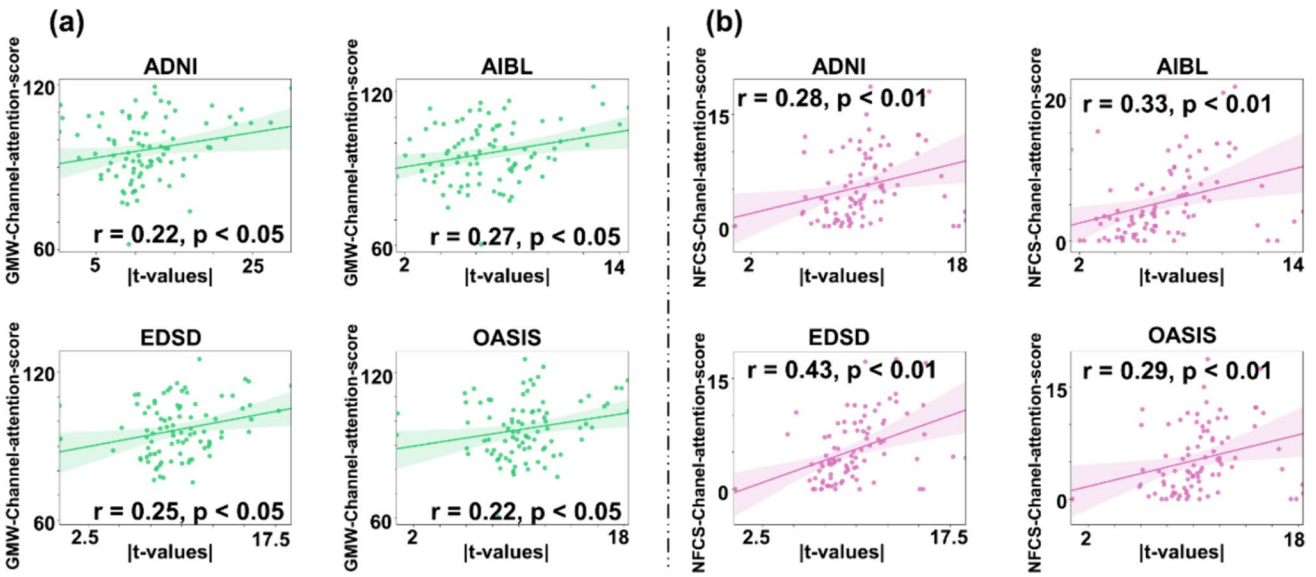


Fig. 4 The correlation scatter plots between the mean attention scores of all the 90 brain regions and the *t*-values (*t* statistic score) on the four databases of ADNI, AIBL, EDSD, and OASIS obtained by the statistical comparison between NC vs AD for the 90 ROIs. **a** The scatter plots between GMV-Channel-attention-score and *t*-values. **b** The scatter plots between NFCS-Channel-attention-score and *t*-values. *ADNI* the Alzheimer's disease neuroimaging initiative database, *AIBL* Australian imaging biomarkers and lifestyle, *EDSD* the European DTI study on dementia database (EDSD), *OASIS* the open access series of imaging studies

The classification score of s²MRI-ADNet was significantly associated with clinical measures

The biological basis of the dual-channel-based deep learning model s²MRI-ADNet was validated by performing a correlation operation between the classification scores generated by the s²MRI-ADNet and the clinical measures in the primary discovery database of ADNI. Figure 5a–h demonstrated a significant correlation between the classification score and the cognitive ability scores (MMSE, ADAS11, ADAS13, ADASQ4, CDRSB, FAQ, Ravlt-immediate (AVLT-1), and Ravlt-learning (AVLT-2)) ($p < 0.0001$ for all), demonstrating a significant association between the s²MRI-ADNet's ability to diagnose AD and cognitive impairment. Besides, the significant correlation was also observed between the classification score and the neuropathological profiles (CSF Aβ, Tau, P-Tau in Fig. 5i–k, $p < 0.001$ for all). Regarding the metabolism and genetic factors, the classification score obtained by the s²MRI-ADNet was also significant correlated with metabolic (FDG) and genetic risk scores (PHS) (Fig. 5l–m, $p < 0.0001$ for all). Those results indicated that our method had solid biological basis for early diagnosis of AD.

Discussion

In this study, an interpretable deep learning model s²MRI-ADNet that integrated multi-dimensional representations of GMV from Euclidean space and the R2SN with a non-Euclidean structure solely from T1WI image was proposed for early diagnosis of AD. The most benefit of the study was adopting one simple and most clinically accessible MR imaging modality to enhance its translation and popularization in a large of number of hospitals. To the best of our knowledge, it is the first attempt to integrate GMV from Euclidean space and the R2SN with a non-Euclidean structure solely from sMRI into a unified framework.

Other than the experiments displayed in the result section (Tables 2, 3), we also summarized number of studies related to early diagnosis of AD in Supplementary Table S2 since we could not collect the same data for comparison particularly for the advanced MRI sequences or in-house

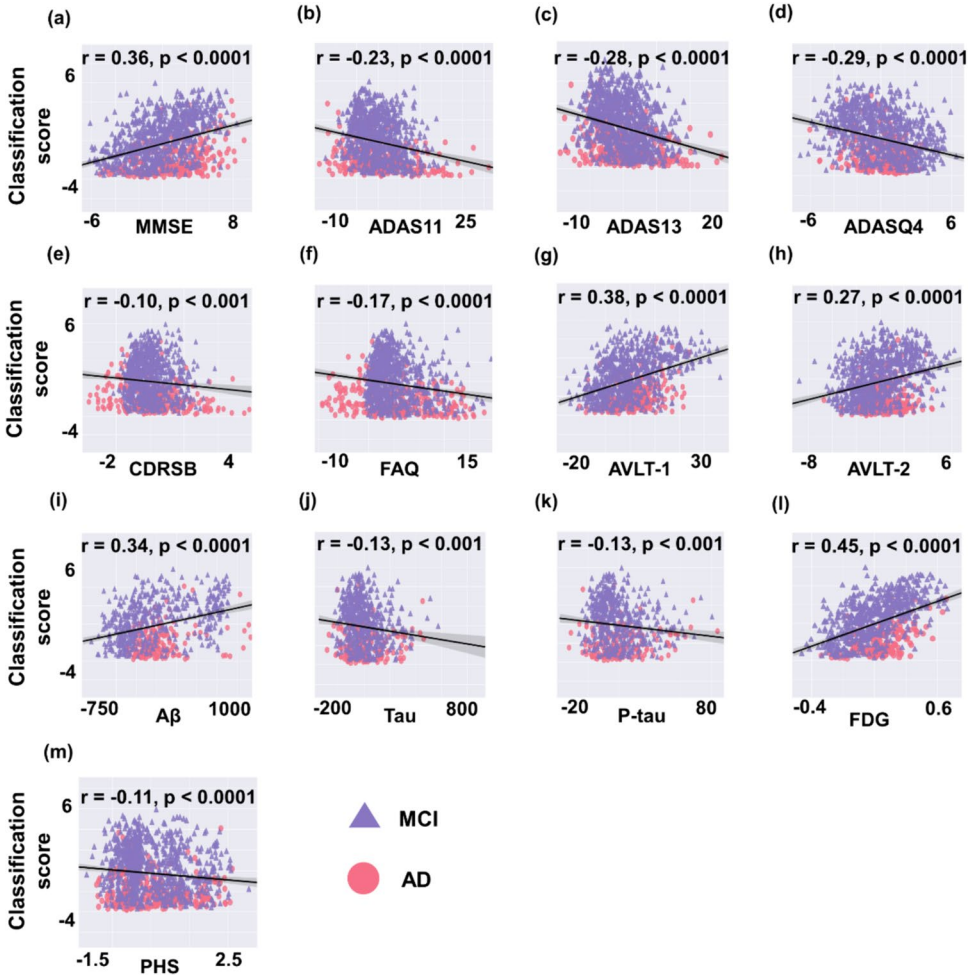


Fig. 5 Scatter plots between classification score and clinical profiles in MCI and AD groups in ADNI cohort. The clinical measures included **a** MMSE, **b** ADAS11, **c** ADAS13, **d** ADASQ4, **e** CDRSB, **f** FAQ, **g** Ravlt-immediate, **h** Ravlt-learning, **i** Aβ, **j** Tau, **k** P-tau, **l** FDG, **m** PHS

data. Supplementary Table S2 illustrates that although s^{2}MRI-ADNet established solely on the sMRI, it still achieved competitive performance in AD analysis by devising the multi-dimensional representations of AD to leverage the heterogeneous features of GMV from Euclidean space and the R2SN with a non-Euclidean structure. Of note, the studies [36, 37] also achieved a relatively superior performance of 93.2% and 93.0% regarding the AD vs NC classification. Specifically, Kong et al. [36] used 3D convolutional neural networks to evaluate an image fusion method, which fused MRI with Positron Emission Tomography (PET) images from AD patients. Pan et al. [37] proposed a novel Hypergraph Generative Adversarial Networks (HGGAN) to generate multimodal connectivity of Brain Network from resting-state functional magnetic resonance imaging (rs-fMRI) combination with Diffusion Tensor Imaging (DTI). However, due to the multi-modality setting, the scale of the experimental data was not sufficient; therefore, the robustness, reproducibility, and generalizability might be limited.

The interpretation of the deep learning model is also important in clinical settings [38], and has attracted increased attentions. Qiu et al. proposed to establish high resolution maps of disease probability in AD analysis [39], Jin et al. employed an attention mechanism module to advance the diagnosis of AD [29]. However, these studies were interpreted only with respect to single input feature. In this study, the interpretation of the deep learning model of s^{2}MRI-ADNet was carried out from two aspects: (1) the complementarity, generalizability, reproducibility of the interpretation of the multi-dimensional representation of AD (Fig. 3); (2) the neuroscientific interpretation (Figs. 4, 5).

From the Fig. 3a, b, our study validated a complementary interpretation of the GMV-Channel and NFCS-Channel in the s^{2}MRI-ADNet. The Transformer encoder in the GMV-Channel primarily captured the brain regions of olfactory cortex, insula, cingulate gyrus, hippocampus, parahippocampus gyrus, amygdala, lingual gyrus, fusiform gyrus, caudate nucleus, lenticular nucleus, thalamus, transverse temporal gyrus (Table 4), while the GNN encoder in the NFCS-Channel primarily captured the brain regions of the occipital lobe (lingual gyrus, lingual gyrus, inferior occipital gyrus, fusiform gyrus), the superior parietal (angular gyrus, precuneus, paracentral lobule) and the temporal lobe (superior temporal gyrus, middle temporal gyrus, inferior temporal gyrus) (Table 4). These regions have been reported to be associated with process of AD in several studies [40, 41]. The complementary characteristics of the dual-channel setting explained the performance improvement of the s^{2}MRI-ADNet in AD analysis.

The generalizability and reproducibility of the interpretation of the s^{2}MRI-ADNet, rather than the sole interpretation, are important for potential clinical translation. In Fig. 3c, we validated the generalizability and reproducibility of the interpretation of the GMV-Channel-attention-map by calculating a significant correlation on four independent databases of ADNI, AIBL, EDSD, and OASIS. Likewise, the Fig. 3d demonstrated the same generalizability and reproducibility on the NFCS-Channel-attention-map.

Besides, the neurobiological interpretation was also validated in terms of brain abnormality and clinical measures. Specifically, we verified the consistency between the attention maps of s^{2}MRI-ADNet and the brain abnormality in Fig. 4, and validated the biological basis of the s^{2}MRI-ADNet by correlating the classification score with mount of clinical measures include including cognitive ability (MMSE, ADAS11, ADAS13, ADASQ4, CDRSB, FAQ, Ravlt-immediate (AVLT-1) and Ravlt-learning (AVLT-2)), CSF biomarker (CSF Aβ, CSF Tau, and CSF P-Tau), metabolism (FDG), and genetic risk score (PHS). Both demonstrated solid neurobiological basis of s^{2}MRI-ADNet in AD analysis.

There are several limitations worthy of being considered in the present study. Although the proposed model has achieved promising performance, our study still needs to be further validated on "real-world" data other than the publicly available database. However, the sMRI is relatively robust for different MRI parameters. Thus, the different MRI parameters are only with limited influence to results. Particularly, our study employed a total of four large independent databases where each cohort was consisted of multiple distinct local sites, such as the ADNI comprising more than 50 sites. Amount of the published studies on ADNI dataset also demonstrated the limited influence caused by different parameters of sMRI such as [42]. Despite various imaging parameters, each T1WI image is normalized and standardized to MNI152 space as an essential preprocessing step in number of studies [29, 43, 44], which actually alleviates the difference between different scans under various imaging parameters. The normalization in the pre-processing step was also adopted in our study. Besides, based on a comprehensive literature review, the publicly available datasets of ADNI [45], AIBL [46], EDSD [47], and OASIS3 [48] covered a wide range of imaging parameters from a variety of sites or MRI scanners. Another four representative studies [29, 49–51] which adopted real-world data from local sites also exhibited a wide range of imaging parameters from a variety of sites or MRI scanners. The imaging parameters are widely overlapped with each other between publicly available datasets and read-world datasets. The various imaging parameters, MRI scanners etc. also demonstrated the robustness of our deep learning model in feature representation of AD and its early diagnosis application.

Conclusion

In summary, an interpretable deep learning model s^{2}MRI-ADNet integrating multi-dimensional representations of AD in Euclidean and graph spaces solely from sMRI was proposed for AD analysis. Since the conventional T1WI image was the most clinically accessible MRI imaging modality, the interpretable deep learning model of s^{2}MRI-ADNet indicated a high potential in clinical translation and popularization.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10334-024-01178-3.

Author contributions Zhiwei Song, Honglun Li and Qiang Zheng conceptualized and designed the study. Zhiwei Song and Yiyu Zhang acquired the data. Zhiwei Song, Honglun Li, Chuanzhen Zhu and Minbo Jiang analyzed and interpreted the data. Limei Song, Yi Wang, Minhui Ouyang, Fang Hu and Qiang Zheng drafted the manuscript. Zhiwei Song, Minhui Ouyang, Fang, and Qiang Zheng critically reviewed the manuscript.

Funding This work was supported by the National Natural Science Foundation of China [61802330, 61802331], Open Project of Key Laboratory of Medical Imaging and Artificial Intelligence of Hunan Province, Xiangnan University [YXZN2022002] and Yantai City Science and Technology Innovation Development Plan [2023XDRH006].

Data availability Experimental data that support the findings of this study are available at the website provided for each database: ADNI (http://adni.loni.usc.edu), AIBL (http://aibl.csiro.au), EDSD (http://neugrid4you.eu), and OASIS (http://oasis-brains.org). Additional data related to this paper may be requested from the authors upon reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

Ethical approval Not applicable.

References

1. Querfurth HW, Laferla FM (2010) Alzheimer's disease. N Engl J Med 362(4):329
2. Logothetis NK (2008) What we can do and what we cannot do with fMRI. Nature 453(7197):869–878
3. Oishi K, Mielke MM, Albert M, Lyketsos CG, Mori S (2011) DTI analyses and clinical applications in Alzheimer's disease. J Alzheimers Dis 26(s3):287–296
4. Zhang N, Gordon ML, Goldberg TE (2017) Cerebral blood flow measured by arterial spin labeling MRI at resting state in normal aging and Alzheimer's disease. Neurosci Biobehav Rev 72:168–175
5. Liu M, Li F, Yan H, Wang K, Ma Y, Shen L, Xu M, AsDN I (2020) A multi-model deep convolutional neural network for automatic hippocampus segmentation and classification in Alzheimer's disease. Neuroimage 208:116459
6. Logothetis NK, Pauls J, Augath M, Trinath T, Oeltermann A (2001) Neurophysiological investigation of the basis of the fMRI signal. Nature 412(6843):150–157
7. Zhao K, Ding Y, Han Y, Fan Y, Alexander-Bloch AF, Han T, Jin D, Liu B, Lu J, Song C (2020) Independent and reproducible hippocampal radiomic biomarkers for multisite Alzheimer's disease: diagnosis, longitudinal progress and biological basis. Sci Bull 65(13):1103–1113
8. Zhang Y, Teng Q, Liu Y, Liu Y, He X (2022) Diagnosis of Alzheimer's disease based on regional attention with sMRI gray matter slices. J Neurosci Method 365:109376
9. Han K, Wang Y, Chen H, Chen X, Guo J, Liu Z, Tang Y, Xiao A, Xu C, Xu Y (2022) A survey on vision transformer. IEEE Trans Pattern Anal Mach Intell 45(1):87–110
10. Dai W, Zhang Z, Tian L, Yu S, Wang S, Dong Z, Zheng H (2022) BrainFormer: a hybrid CNN-transformer model for brain fMRI data classification. arXiv preprint arXiv:220803028.
11. Alexander-Bloch A, Giedd JN, Bullmore E (2013) Imaging structural co-variance between human brain regions. Nat Rev Neurosci 14(5):322–336
12. Zhao K, Zheng Q, Che T, Dyrba M, Li Q, Ding Y, Zheng Y, Liu Y, Li S (2021) Regional radiomics similarity networks (R2SNs) in the human brain: reproducibility, small-world properties and a biological basis. Netw Neurosci 5(3):783–797
13. Zhao K, Zheng Q, Dyrba M, Rittman T, Li A, Che T, Chen P, Sun Y, Kang X, Li Q (2022) Regional radiomics similarity networks reveal distinct subtypes and abnormality patterns in mild cognitive impairment. Adv Sci 9(12):2104538
14. Yao D, Sui J, Yang E, Yap P, Shen D, Liu M (2020) Temporal-adaptive graph convolutional network for automated identification of major depressive disorder using resting-state fMRI. International workshop on machine learning in medical imaging. Springer, New York, pp 1–10
15. Yao D, Sui J, Wang M, Yang E, Jiaerken Y, Luo N, Yap P-T, Liu M, Shen D (2021) A mutual multi-scale triplet graph convolutional network for classification of brain disorders using functional or structural connectivity. IEEE Trans Med Imaging 40(4):1279–1289
16. Wang J, Knol MJ, Tiulpin A, Dubost F, de Bruijne M, Vernooij MW, Adams HH, Ikram MA, Niessen WJ, Roshchupkin GV (2019) Gray matter age prediction as a biomarker for risk of dementia. Proc Natl Acad Sci 116(42):21213–21218
17. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 770–778
18. Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G, Gelly S (2020) An image is worth 16x16 words: transformers for image recognition at scale. arXiv preprint arXiv:201011929
19. Guo J, Han K, Wu H, Tang Y, Chen X, Wang Y, Xu C (2022) Cmt: Convolutional neural networks meet vision transformers. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 12175–12185
20. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I (2017) Attention is all you need. Advances in Neural Information Processing Systems 30
21. Hamilton W, Ying Z, Leskovec J (2017) Inductive representation learning on large graphs. Advances in Neural Information Processing Systems 30
22. Khoshraftar S, An A (2024) A survey on graph representation learning methods. ACM Trans Intell Syst Technol 15(1):1–55
23. Zhao M, Yan W, Luo N, Zhi D, Fu Z, Du Y, Yu S, Jiang T, Calhoun VD, Sui J (2022) An attention-based hybrid deep learning framework integrating brain connectivity and activity of resting-state functional MRI data. Med Image Anal 78:102413

24. Sun L, Peng Q, Qian C, Li J Tau (2022) Content prediction based on brain network MLP-att model. In: 2022 international conference on machine learning, cloud computing and intelligent mining (MLCCIM). IEEE, pp 353–358
25. Zhang S, Chen X, Shen X, Ren B, Yu Z, Yang H, Jiang X, Shen D, Zhou Y, Zhang X-Y (2023) A-GCL: Adversarial graph contrastive learning for fMRI analysis to diagnose neurodevelopmental disorders. Med Image Anal 90:102932
26. Xia M, Wang J, He Y (2013) BrainNet Viewer: a network visualization tool for human brain connectomics. PLoS ONE 8(7):e68910
27. Wang J, Wang X, Xia M, Liao X, Evans A, He Y (2015) GRETNA: a graph theoretical network analysis toolbox for imaging connectomics. Front Hum Neurosci 9:386
28. Zhang Y, Li H, Zheng Q (2023) A comprehensive characterization of hippocampal feature ensemble serves as individualized brain signature for Alzheimer's disease: deep learning analysis in 3238 participants worldwide. Eur Radiol 33(8):5385–5397
29. Jin D, Zhou B, Han Y, Ren J, Han T, Liu B, Lu J, Song C, Wang P, Wang D (2020) Generalizable, reproducible, and neuroscientifically interpretable imaging biomarkers for Alzheimer's disease. Adv Sci 7(14):2000675
30. Ebrahimi A, Luo S, Chiong R (2020) Introducing transfer learning to 3D ResNet-18 for Alzheimer's disease detection on MRI images. In: 2020 35th international conference on image and vision computing New Zealand (IVCNZ), Wellington, New Zealand. IEEE, pp 1–6
31. AI-Tam RM, AI-Hejri AM, Narangale SM, Samee NA, Mahmoud NF, AI-Masni MA, AI-Antari MA (2022) A hybrid workflow of residual convolutional transformer encoder for breast cancer classificationi using digital X-ray mammograms. Biomedicines 10(11):1971
32. Kipf TN, Welling M (2016) Semi-supervised classification with graph convolutional networks. arXiv preprint arXiv:160902907
33. Taud H, Mas J (2018) Multilayer perceptron (MLP). In: Geomatic approaches for modeling land change scenarios. Springer, pp 451–455
34. Veličković P, Cucurull G, Casanova A, Romero A, Lio P, Bengio Y (2017) Graph attention networks. arXiv preprint arXiv:171010903
35. Hu W, Liu B, Gomes J, Zitnik M, Liang P, Pande V, Leskovec J (2019) Strategies for pre-training graph neural networks. arXiv preprint arXiv:190512265
36. Kong Z, Zhang M, Zhu W, Yi Y, Wang T, Zhang B (2022) Multimodal data Alzheimer's disease detection based on 3D convolution. Biomed Signal Process Control 75:103565
37. Pan J, Lei B, Shen Y, Liu Y, Feng Z, Wang S (2021) Characterization multimodal connectivity of brain network by hypergraph GAN for Alzheimer's disease analysis. Chinese conference on pattern recognition and computer vision (PRCV). Springer, pp 467–478
38. Rasheed K, Qayyum A, Ghaly M, Al-Fuqaha A, Razi A, Qadir J (2022) Explainable, trustworthy, and ethical machine learning for healthcare: a survey. Comput Biol Med 149:106043
39. Qiu S, Joshi PS, Miller MI, Xue C, Zhou X, Karjadi C, Chang GH, Joshi AS, Dwyer B, Zhu S (2020) Development and validation of an interpretable deep learning framework for Alzheimer's disease classification. Brain 143(6):1920–1933
40. Bäckman L, Andersson J, Nyberg L, Winblad B, Nordberg A, Almkvist O (1999) Brain regions associated with episodic retrieval in normal aging and Alzheimer's disease. Neurology 52(9):1861–1861
41. Zhang Y, Wang S, Phillips P, Yang J, Yuan T-F (2016) Three-dimensional eigenbrain for the detection of subjects and brain regions related with Alzheimer's disease. J Alzheimers Dis 50(4):1163–1179
42. Ding Y, Sohn JH, Kawczynski MG, Trivedi H, Harnish R, Jenkins NW, Lituiev D, Copeland TP, Aboian MS, Mari Aparici C (2019) A deep learning model to predict a diagnosis of Alzheimer disease by using 18F-FDG PET of the brain. Radiology 290(2):456–464
43. Caso F, Agosta F, Mattavelli D, Migliaccio R, Canu E, Magnani G, Marcone A, Copetti M, Falautano M, Comi G (2015) White matter degeneration in atypical Alzheimer disease. Radiology 277(1):162–172
44. Möller C, Pijnenburg YA, van der Flier WM, Versteeg A, Tijms B, de Munck JC, Hafkemeijer A, Rombouts SA, van der Grond J, van Swieten J (2016) Alzheimer disease and behavioral variant frontotemporal dementia: automatic classification based on cortical atrophy for single-subject diagnosis. Radiology 279(3):838–848
45. Jack CR, Bernstein MA, Fox NC, Thompson P, Alexander G, Harvey D, Borowski B, BritsonWhitwell PJJL, Ward CJ (2008) The Alzheimer's disease neuroimaging initiative (ADNI): MRI methods. J Magn Reson Imaging 27(4):685–691
46. Rowe CC, Ellis KA, Rimajova M, Bourgeat P, Pike KE, Jones G, Fripp J, Tochon-Danguy H, Morandeau L, O'Keefe G (2010) Amyloid imaging results from the Australian imaging, biomarkers and lifestyle (AIBL) study of aging. Neurobiol Aging 31(8):1275–1283
47. Brueggen K, Grothe MJ, Dyrba M, Fellgiebel A, Fischer F, Filippi M, Agosta F, Nestor P, Meisenzahl E, Blautzik J (2017) The European DTI study on dementia—a multicenter DTI and MRI study on Alzheimer's disease and mild cognitive impairment. Neuroimage 144:305–308
48. Marcus DS, Wang TH, Parker J, Csernansky JG, Morris JC, Buckner RL (2007) Open access series of imaging studies (OASIS): cross-sectional MRI data in young, middle aged, nondemented, and demented older adults. J Cogn Neurosci 19(9):1498–1507
49. Mehta RI, Carpenter JS, Mehta RI, Haut MW, Ranjan M, Najib U, Lockman P, Wang P, D'haese P-F, Rezai AR (2021) Blood-brain barrier opening with MRI-guided focused ultrasound elicits meningeal venous permeability in humans with early Alzheimer disease. Radiology 298(3):654–662
50. Choi JD, Moon Y, Kim H-J, Yim Y, Lee S, Moon W-J (2022) Choroid plexus volume and permeability at brain MRI within the Alzheimer disease clinical spectrum. Radiology 304(3):635–645
51. Damulina A, Pirpamer L, Soellradl M, Sackl M, Tinauer C, Hofer E, Enzinger C, Gesierich B, Duering M, Ropele S (2020) Cross-sectional and longitudinal assessment of brain iron level in Alzheimer disease using 3-T MRI. Radiology 296(3):619–626

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.